

数学软件大作业：地月借力太阳近掠轨道的数值模拟与优化

本文实现 2026 年地月借力进入太阳近掠轨道的 reduced patched-conic 扫描项目。交付物包括 Python 数值代码、测试、图表、动画、README、AI 使用声明和可复现 Makefile。

方法概述

模型使用 AU-year 规范单位，太阳引力参数取 $4\pi^2$ 。太阳近掠段由 vis-viva 公式计算，月球借力段用 SOI、双曲转角和月球公转速度进行透明近似。Velocity-Verlet 用二体圆轨道做能量稳定性验证，Horizons 接口强制使用 @10/@399 形式，并提供 fetch_horizons_vectors 函数实时调用 JPL Horizons HTTP API。

扫描结果摘要

```
launch_date: 2026-05-29
lunar_perilune_km: 50000.0
side: trailing
solar_perihelion_au: 0.4000
launch_delta_v_km_s: 6.1642
flyby_delta_v_km_s: 0.0341
reentry_delta_v_km_s: 0.0000
total_delta_v_km_s: 6.1983
v_inf_km_s: 6.0342
turn_angle_deg: 0.31
total_time_years: 0.3048
lunar_soi_km: 66182.9
```

误差与局限

项目包含解析月球偏折与月心数值飞越的对比验证、最优解附近的发射日期/近月距灵敏度分析、以及 Velocity-Verlet 步长收敛性分析。主扫描仍不是完整高精度任务设计，尚未替代 Lambert 求解、三维 B 平面优化和全年 Horizons 高精度对比。

AI 使用说明

AI 辅助用于项目结构、代码草稿、测试、图表脚本、报告和说明文件。物理假设保留在源码中供人工审查。

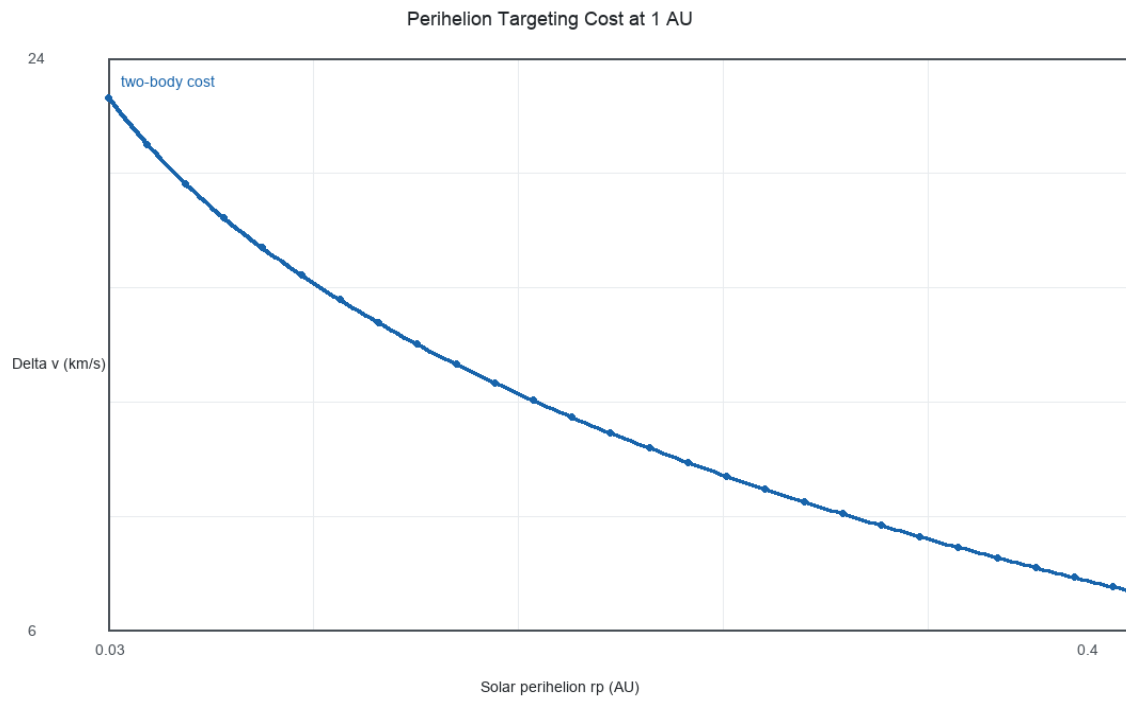


Figure 1. Two-body perihelion targeting cost.

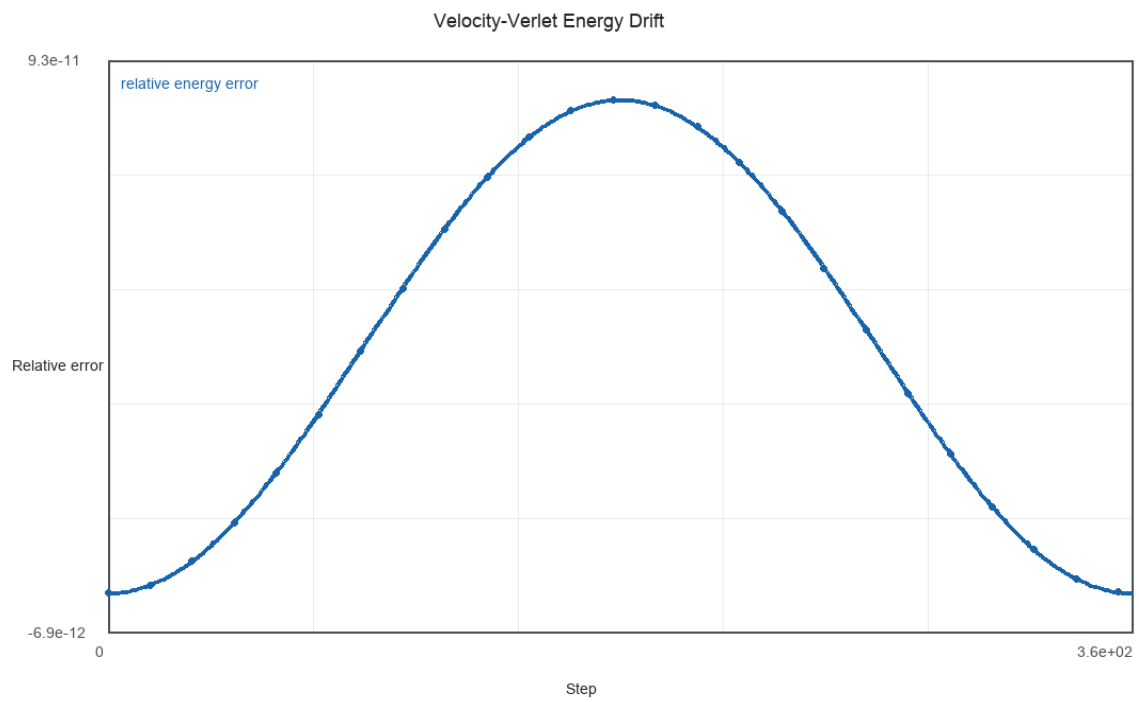


Figure 2. Velocity-Verlet validation.

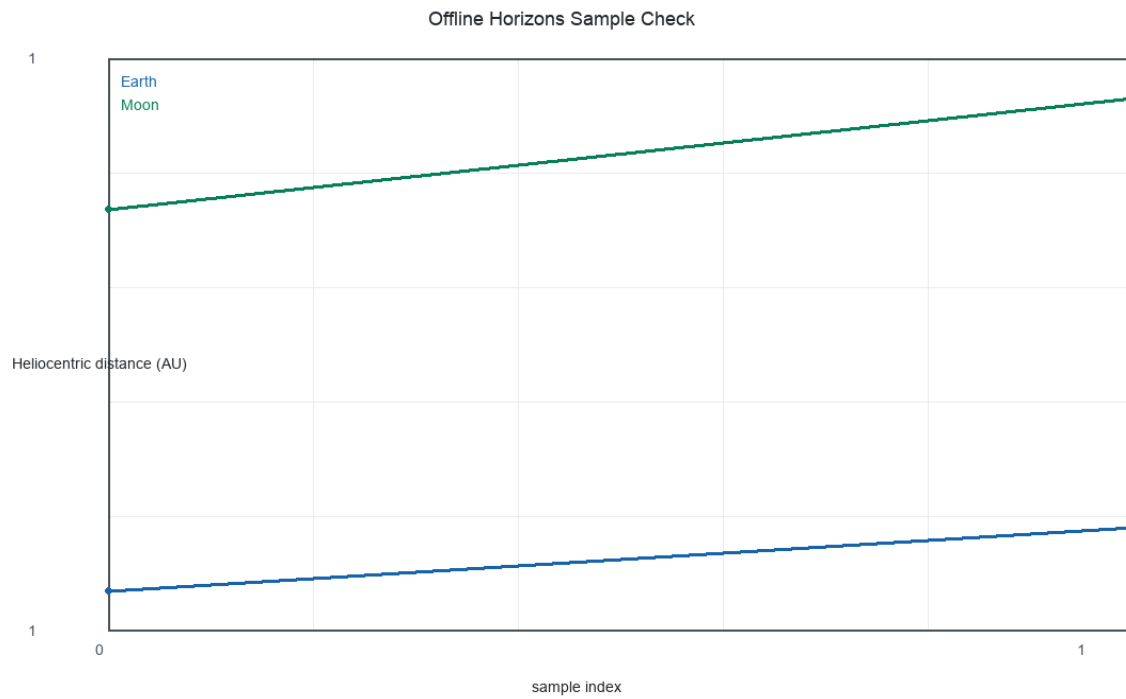


Figure 3. Offline Horizons sample check.

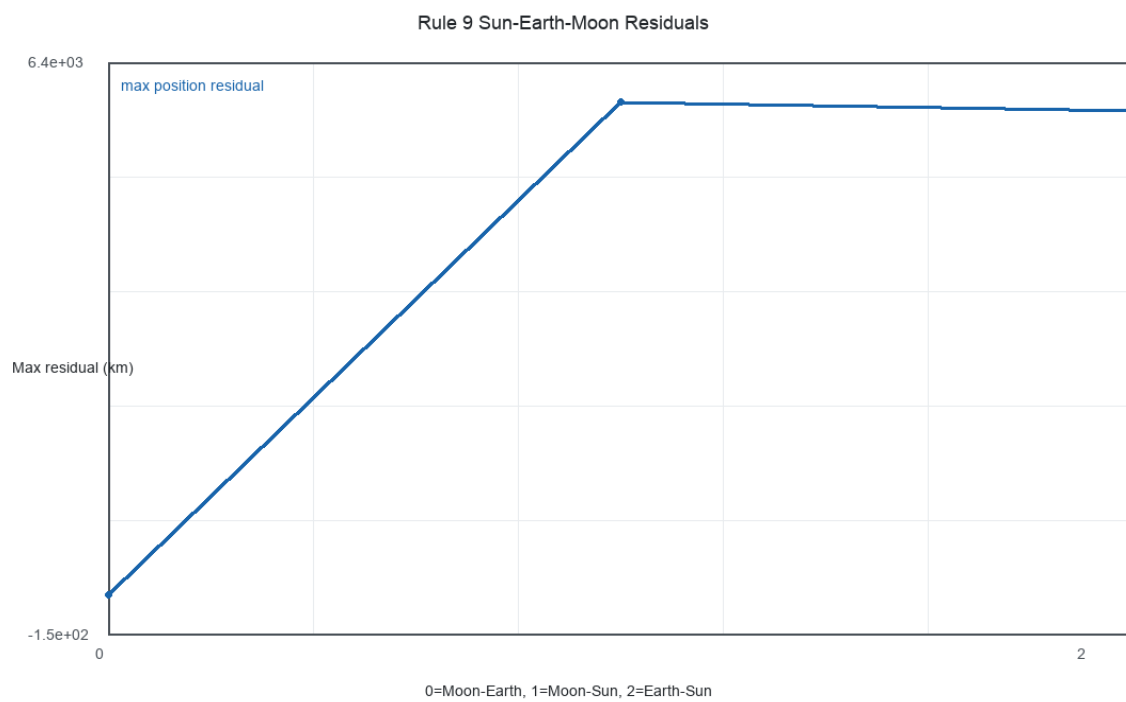


Figure 3b. Rule-9 Horizons residual validation.

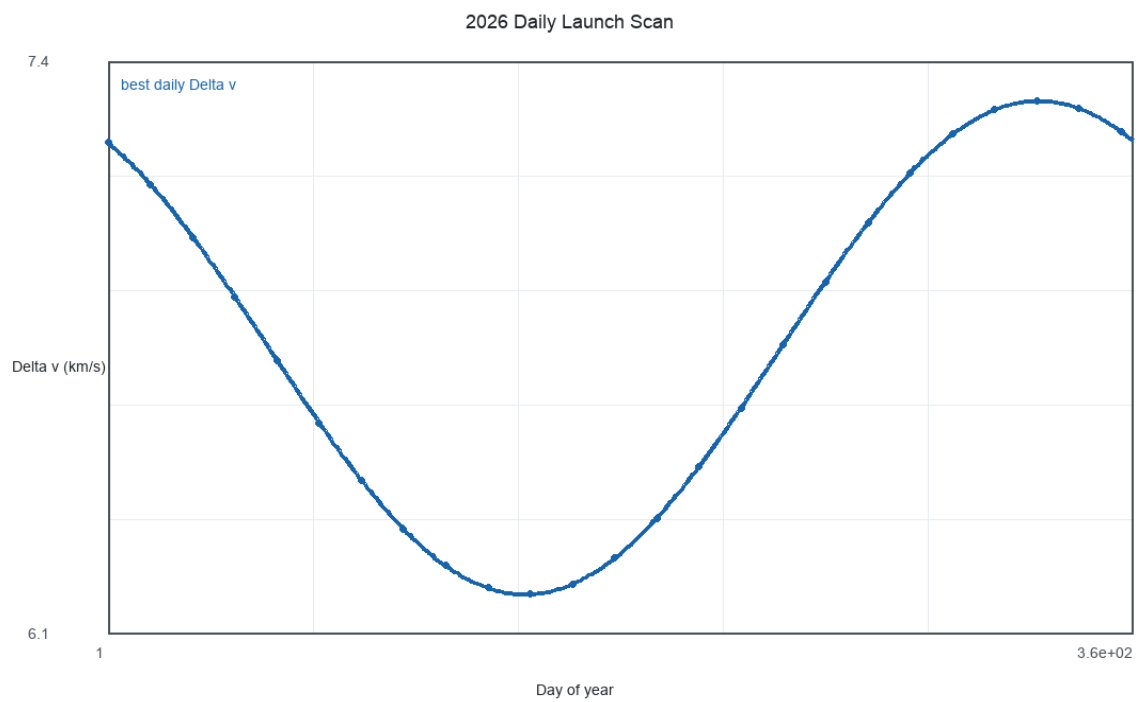


Figure 4. 2026 daily launch scan.

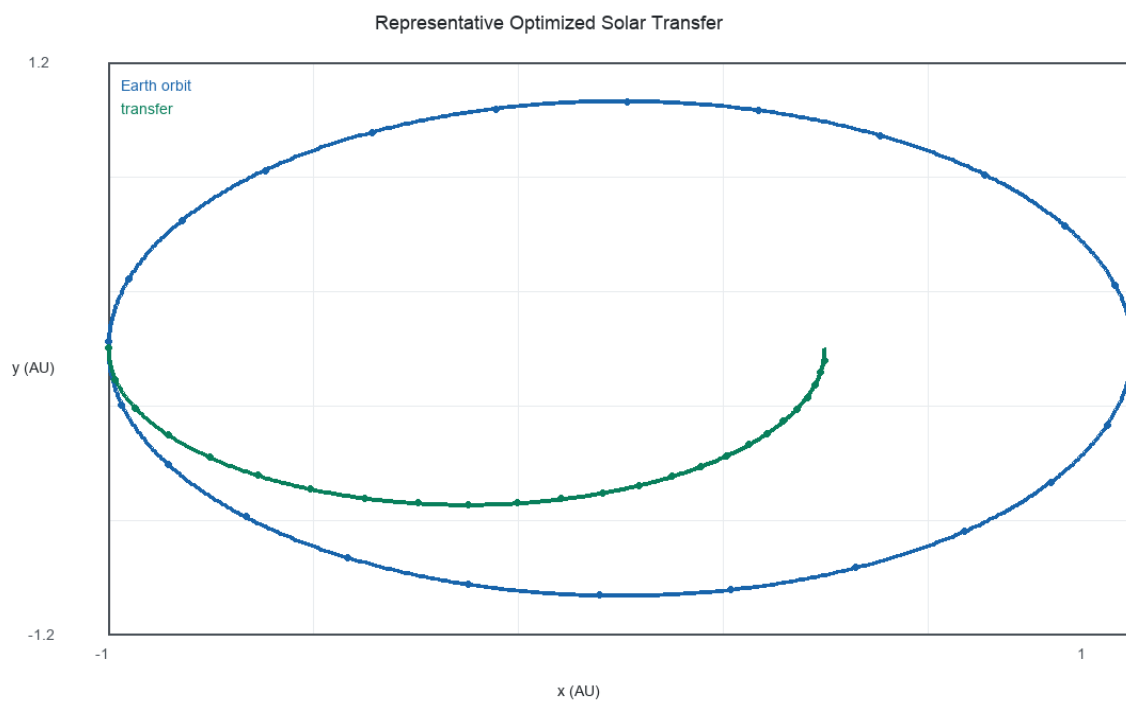


Figure 5. Representative optimized transfer.

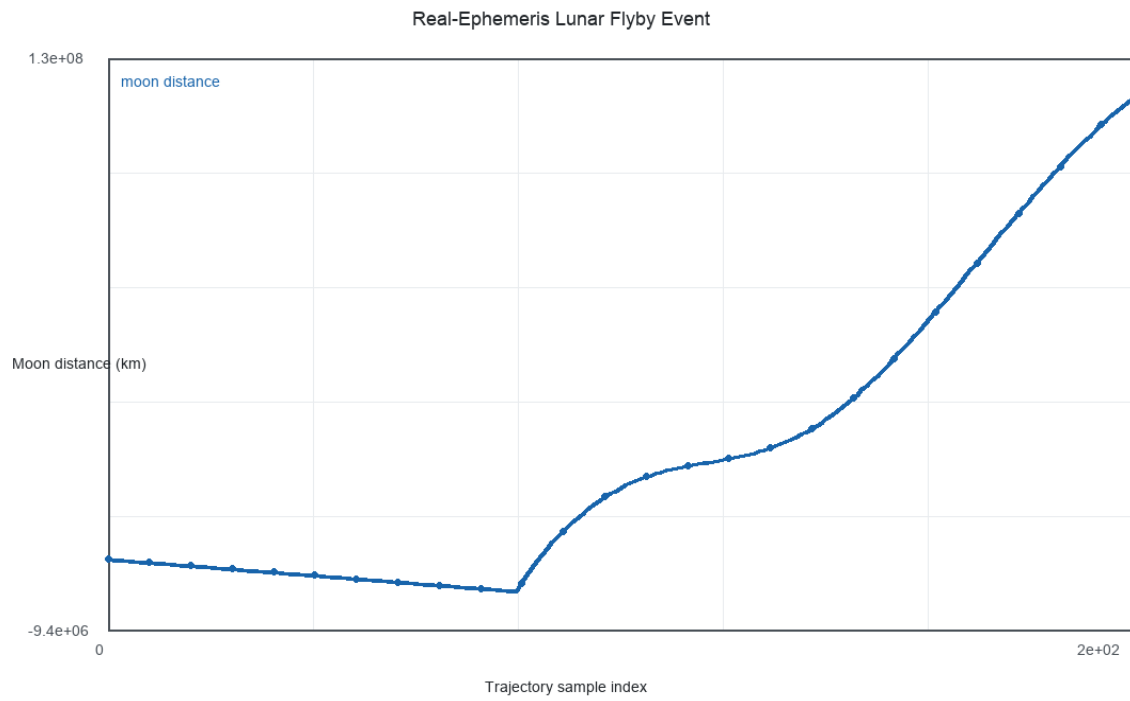


Figure 5b. Real-ephemeris lunar flyby event.

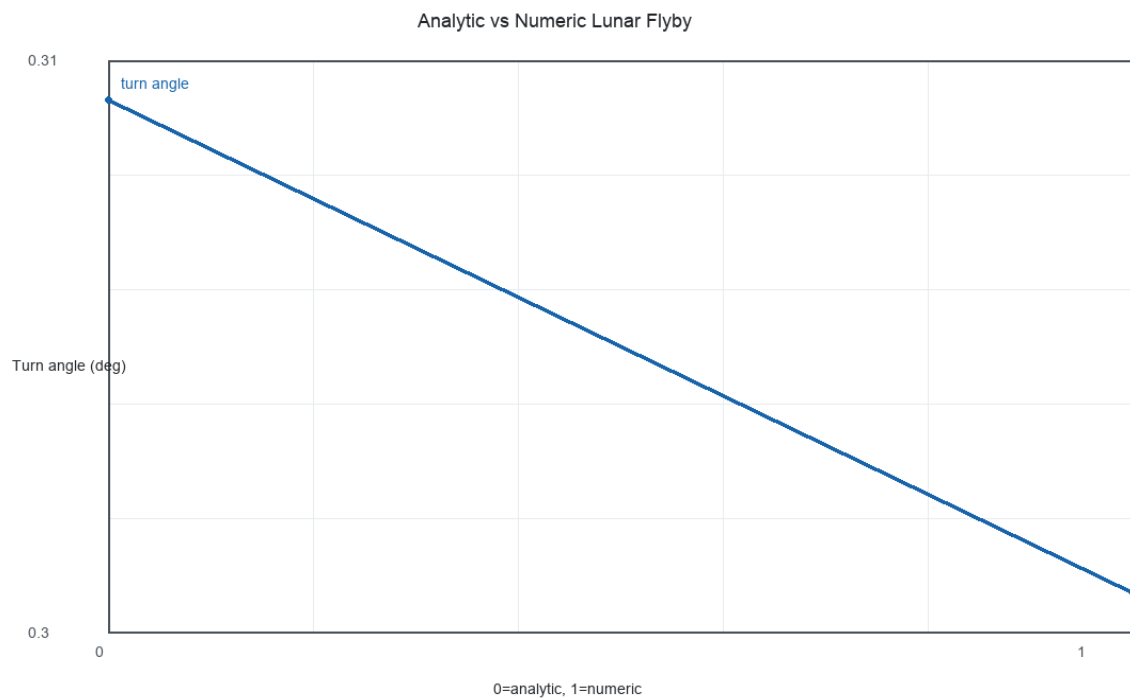


Figure 6. Analytic vs numeric lunar flyby validation.

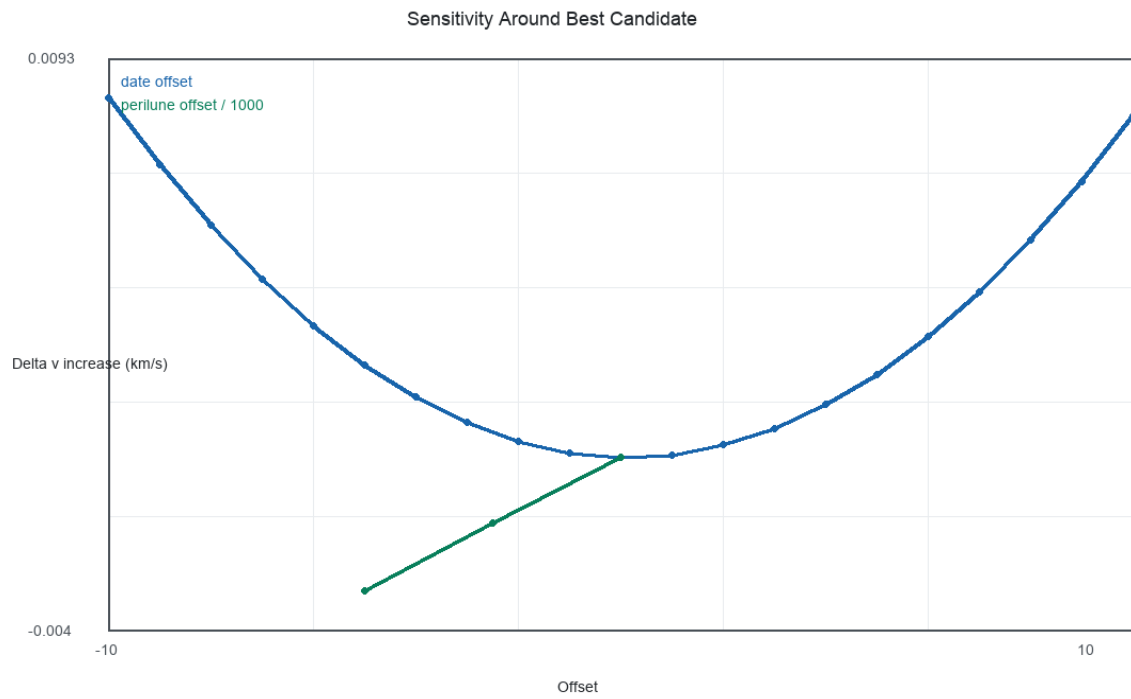


Figure 7. Sensitivity around the best candidate.

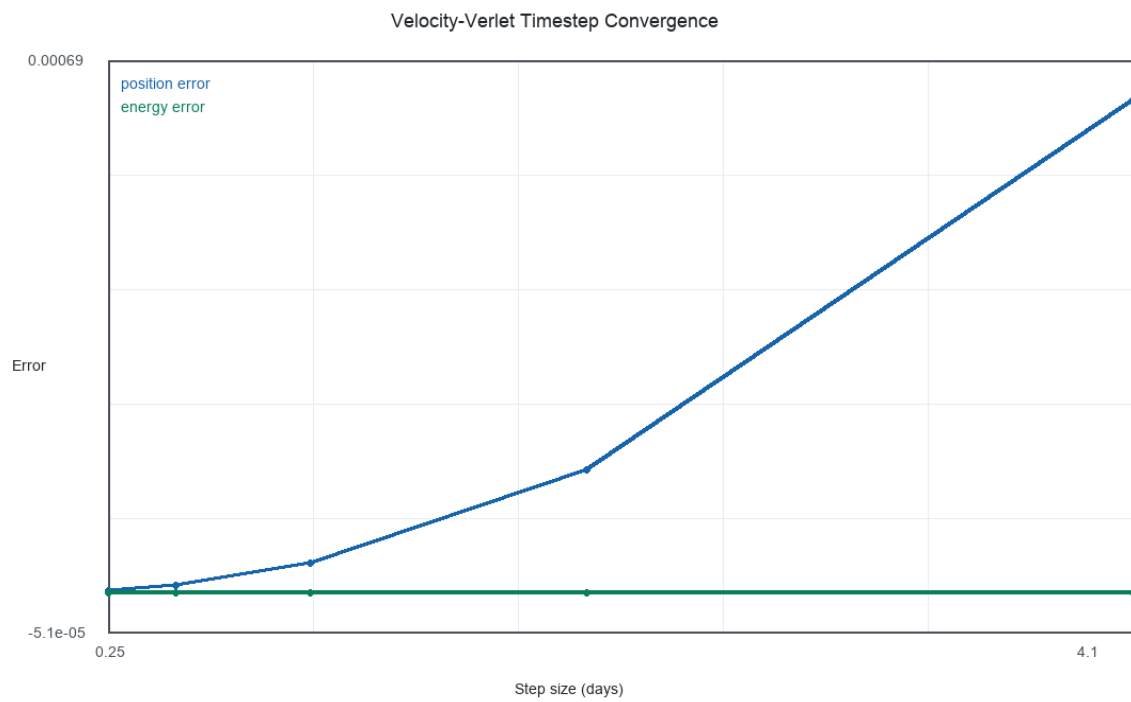


Figure 8. Timestep convergence.

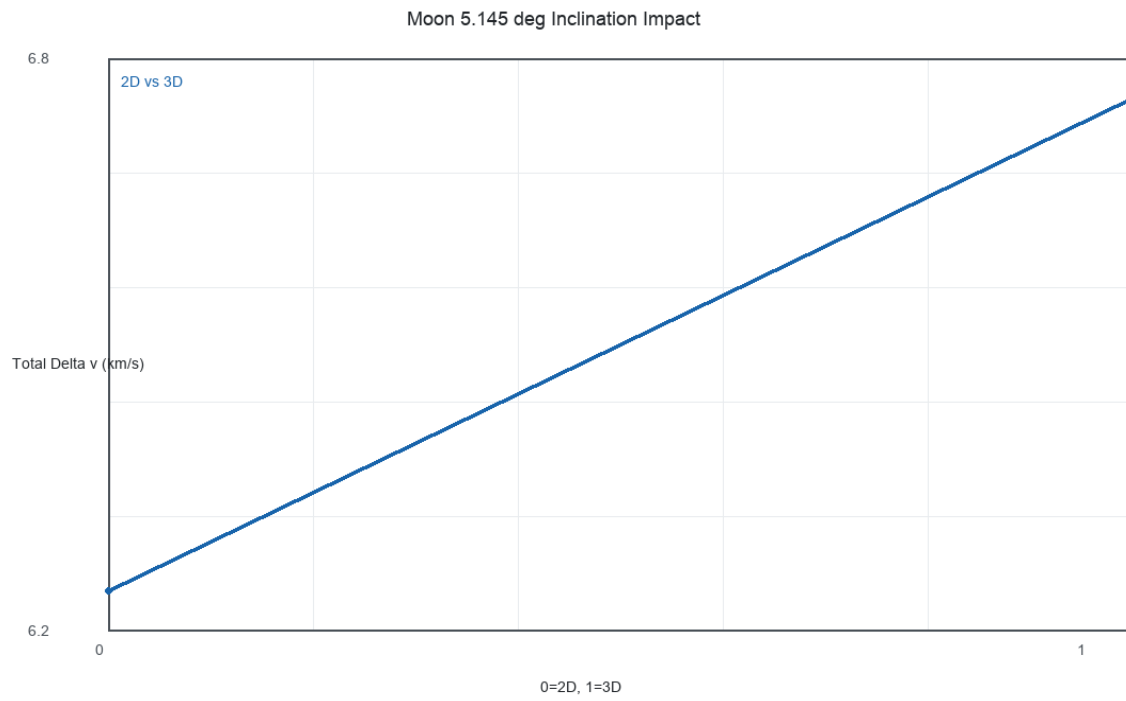


Figure 9. Moon inclination impact.

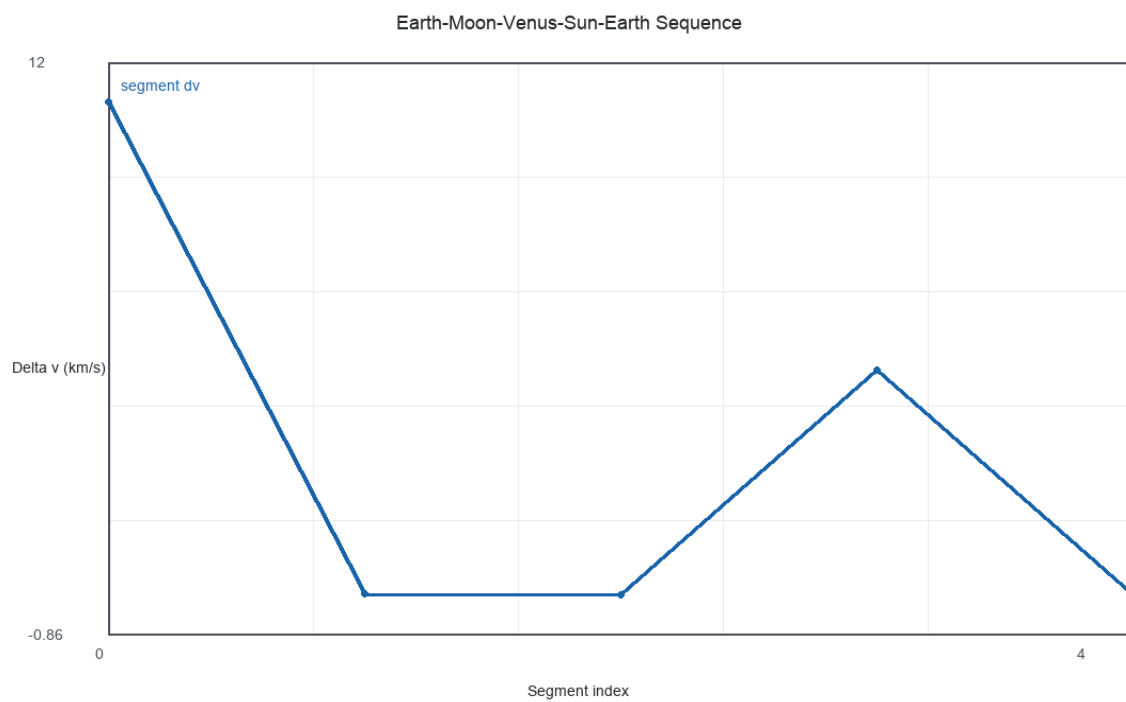


Figure 10. Earth-Moon-Venus-Sun-Earth sequence.

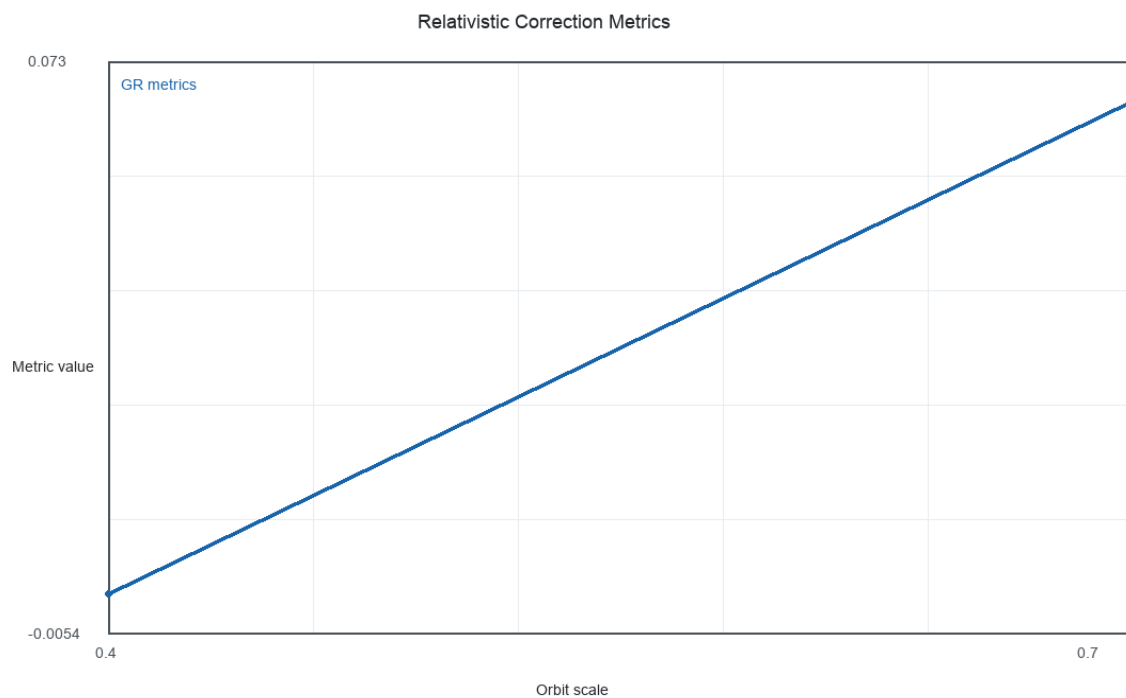


Figure 11. Relativistic correction metrics.

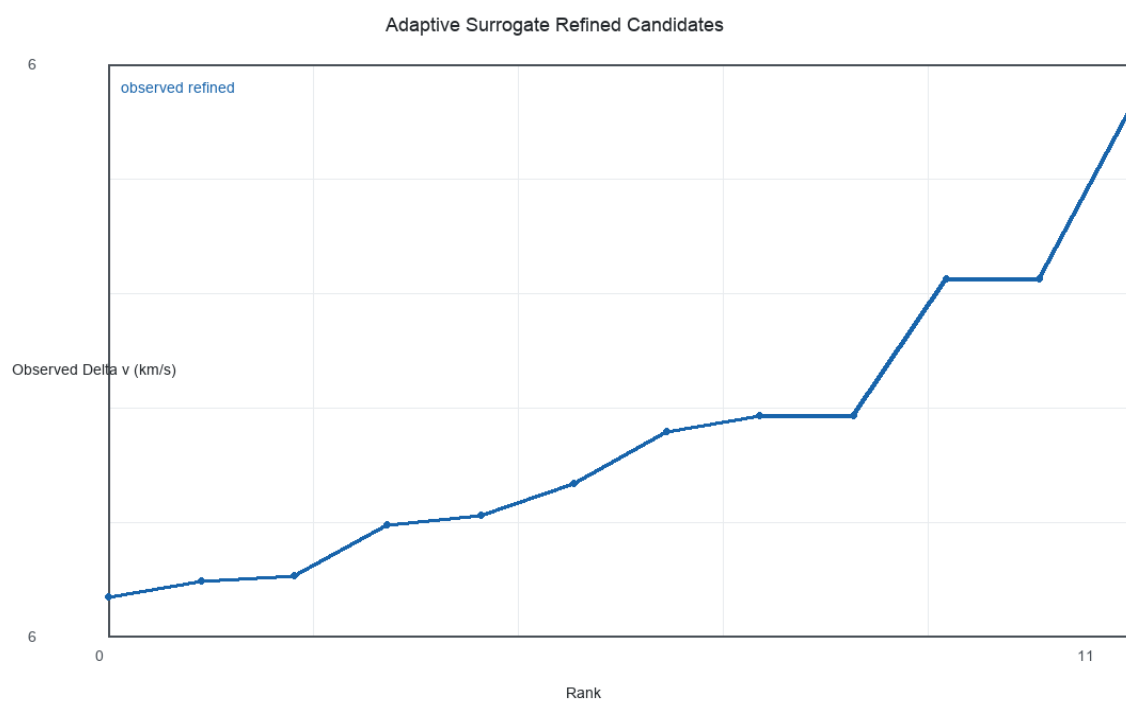


Figure 12. Adaptive surrogate scan.